

AMS / MagEAM — Blueprint Completo del Sistema

Versión: 1.0 · 2026-05-12

Owner: David Cabezas (VSC)

Cliente principal: Cliente Minero CL-SN (anonimizado)

Stack: React 19 + Vite · FastAPI + SQLAlchemy 2.0 · SQLite · Docker · nginx

Deploy: mageam.com (Hetzner VPS) · HEAD cd83670

Estadísticas:

- 57 routers FastAPI
- 86 services Python
- 67 modelos BD
- 562 endpoints API
- 4 agentes IA + 36 skills
- 1593 tests pytest

1. Arquitectura de alto nivel

...

[illegible]

■ USUARIOS (4 roles) ■

■ Operador · Mantenedor · Supervisor · Planner · Programador ■

■ Ing. Confiabilidad · Manager · Admin ■

[illegible]

■ HTTPS

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■ Web App (React) ■ ■ Field App (PWA) ■

■ mageam.com ■ ■ mageam.com/field ■

[illegible]

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[illegible]

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3. Frontend (React + Vite)

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[illegible]

■ PAGES (38 archivos) ■

[REDACTED]

■ ■

■ Dashboard ExecutiveDashboard AnalyticsPage ■

■ FailureCapture FailuresEvents FMECA / FMEA ■

■ WorkOrdersPage WorkRequests WorkManagement ■

■ Planning Scheduling (v2 Horarios) SupervisorBoard ■

■ Execution ExecutionChecklists PostMaintenance ■

■ Criticality Reliability RCA / RagLab ■

■ Backlog Team SapPmPage ■

■ Settings SettingsPage Profile ■

■ AgenticCapabilities AIAgents AuditLogPage ■

■ DataImport DocumentManagement DefectElim. ■

■ ImprovementActions ContractorsPage ProjectSelector ■

■ PerformanceAnalysis ExpertKnowledge FieldCapture ■

■ ProgrammerAgent SupervisorAgent PlanificadorAg. ■

[illegible]

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[illegible]

■ COMPONENTES COMPARTIDOS ■

■ Header · Sidebar · WelcomeCard · Toast · ConfirmDialog ■

■ SmartCaptureModal · ExecutiveView · TacticalOperationsView ■

■ PageGuard (RBAC) · WsStatusIndicator · PriorityBadge ■

■ KPICard · CancelWOModal · SmartAssignModal ■

[illegible]

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■ Backend: capture_service ■

■ → comprime foto ■

■ → guarda en data/capture_* ■

■ → AI suggest (Claude Vision) ■

■ → crea WorkRequest (DRAFT) ■

[illegible]

▼ AV-NNNNN code

[illegible]

■ SUPERVISOR aprueba ■

■ PUT /work-requests/{id}/ ■

■ validate {APPROVE} ■

■ → WR.status = APROBADO ■

■ → backlog_service.add() ■

XX

■

▼ BacklogItem

XX

■ PLANNER convierte ■

■ POST /managed-work-orders/ ■

■ from-wr ■

■ → OT-2026-NNNNN ■

■ → status=CREADO ■

XX

■

▼ ManagedWorkOrder (wo_id)

XX

■ Walk-through estados: ■

■ CREADO → LIBERADO → ■

■ PLANIFICADO → ■

■ EN PROGRAMACION → ■

■ PROGRAMADO → ■

■ EN_EJECUCION → CERRADO ■

□ □

■ Cada transición: ■

■ - audit_log entry ■

■ - WS broadcast wo_updated ■

...

6. RAG (Retrieval Augmented Generation)

...

[REDACTED]

■ LanceDB (data/lancedb/) ■

■ ■

■ Tables: ■

■ - ot_history (56 OTs cerradas indexed) ■

■ - lessons_learned (post-cierre learnings) ■

■ - manuals (DMS manuals + procedures) ■

■ ■

■ Each row: {vector(384), text, source_id, source_type, meta} ■

[illegible]

- Embed model:

■ paraphrase-multilingual-MiniLM-L12-v2

■ (sentence-transformers)

■ 1.5GB RAM al primer uso

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[REDACTED]

■ rag_service ■

11

■ add_chunks(table, items) — ingesta con chunking 500 tokens ■

■ `search(table, query, k=5)` — similarity search ■

■ `embed(texts)` — vectoriza (cached) ■

■ stats() — health check (loaded: bool) ■

[illegible]

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▼ ▼

[illegible]

[illegible]

7. Permission Matrix (RBAC)

■ 6 ROLES ■

- admin — all permissions ■
- manager — executive view ■
- planner — planning scope ■
- engineer — reliability scope ■
- supervisor — execution + own ■
- tecnico — execution + own ■

■

▼

■ 24 MODULES × 2 ACTIONS ■

- (view, edit) ■
- ■

■ PERMISSIONS dict in ■

- usePermissions.js (frontend) ■
- + require_role() decorators ■
- (backend) ■

■

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...

...

[illegible]

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...

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■ ports: 80/443 ■ ■ port 8000 ■ ■ port 5173 ■

[illegible][illegible]

```

■ db data ■ ■ lancedb/ ■ ■ hf cache ■

```

■ audit) ■ ■ ■ ■ model) ■

[illegible]

...

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```
■ git push origin main ■ ■
```

[illegible]

■ ■ GitHub Actions ■

■ ■ - pytest smoke critical ■

■ ■ - npm run build ■

■ ■ ■

■ (ssh remote bare repo) ■ ■

12345678910111213141516171819202122232425262728293031323334353637383940414243444546474849505152535455565758596061626364656667686970717273747576777879808182838485868788899091929394959697989910010110210310410510610710810911011111211311411511611711811912012112212312412512612712812913013113213313413513613713813914014114214314414514614714814915015115215315415515615715815916016116216316416516616716816917017117217317417517617717817918018118218318418518618718818919019119219319419519619719819920020120220320420520620720820921021121221321421521621721821922022122222322422522622722822923023123223323423523623723823924024124224324424524624724824925025125225325425525625725825926026126226326426526626726826927027127227327427527627727827928028128228328428528628728828929029129229329429529629729829930030130230330430530630730830931031131231331431531631731831932032132232332432532632732832933033133233333433533633733833934034134234334434534634734834935035135235335435535635735835936036136236336436536636736836937037137237337437537637737837938038138238338438538638738838939039139239339439539639739839940040140240340440540640740840941041141241341441541641741841942042142242342442542642742842943043143243343443543643743843944044144244344444544644744844945045145245345445545645745845946046146246346446546646746846947047147247347447547647747847948048148248348448548648748848949049149249349449549649749849950050150250350450550650750850951051151251351451551651751851952052152

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```
■ ssh root@ ■
```

- ocp-backend ocp-frontend ■
- docker compose up -d ■
- docker compose restart ■
- ocp-nginx ■
- ■
- (15-20 segundos) ■
- ■
- Service health checks ■
- /api/v1/health ■
- ...

11. Decisiones de diseño claves

Decisión	Por qué
JSON embebido en `operations/materials/workers`	Snapshot inmutable + flexibilidad schema cliente
Sin FK formal en `work_request_id`	OTs fast-track P1 sin WR previo + robustez a borrados
Audit log universal (entity_type + id)	1 tabla vs N tablas, query simple cross-entity
4 agentes específicos vs 1 mega-agente	Especialización + token budget separado por rol
LanceDB embedded vs vector DB externo	0 ops overhead, RAG funciona sin servicios extra
SQLite vs Postgres	Single-tenant inicial, deploy simple; migrar fácil
Optimistic locking (version + If-Match)	UX sin lock contention vs pessimistic
Stub queue para SAP	Cliente sin credenciales aún; pipeline listo
Lazy load embed model	Idle 0 RAM; 1.5GB solo al primer query
Scoped roles via JWT plant_id	IDOR-mitigation desde JWT, no query param

12. Hitos completados (commit reciente)

Sprint	Tickets cerrados	Highlights
SP7 VSC (en curso)	20 cerrables, 2 parciales	SF-656 audit · SF-660 policy · SF-661 v0.2 IA · SF-662 preparativos · 0A2 Gap 95% · Tutorial v2 · 12 fixes pytest
SP6 VSC	Audit BD · multi-plant · IDOR fixes	seed criticality · 8 workforce · DMS auth
SP5 VSC	Field PWA · capture flow · planner agent	offline-first · audio transcription · IA suggest

13. Lo que viene (SP8 — backlog)

- **SF-661 funciones 2/4/7** — req histórico real (0B2)

- **SF-662 UI completa** — req spec final Jorge
 - **0E centros de costo** + clases de gasto
 - **0G1 casos de uso Carlos** — input para E2E Playwright
 - **Refactor módulo Ejecución** (deadline VIE 15-may)
 - **vitest + Playwright CI nightly**
 - **S3/blob storage para fotos** (hoy local FS)
 - **SAP integration real** cuando lleguen credenciales
 - **Análisis Desempeño** granularidad por work_center
-

14. Referencias

- [docs/MODEL_RELATIONSHIPS.pdf](MODEL_RELATIONSHIPS.pdf) — grafo de relaciones OT
 - [docs/AUDIT_LOG_POLICY.pdf](AUDIT_LOG_POLICY.pdf) — política trazabilidad
 - [docs/OT_CALENDAR_SYNC.pdf](OT_CALENDAR_SYNC.pdf) — sync OT-calendario
 - [docs/TANDAS_SP7_SP8_2026-05-11.pdf](TANDAS_SP7_SP8_2026-05-11.pdf) — roadmap
 - [1-output/M3/gap-analysis-magdalena-2026-05-12.xlsx](../1-output/M3/gap-analysis-magdalena-2026-05-12.xlsx) — cruce 174 items vs código
 - [1-output/M3/testing-strategy-magam-2026-05-12.pdf](../1-output/M3/testing-strategy-magam-2026-05-12.pdf) — pirámide testing
 - [docs/AMSA_BBP_PM_04_Rev_0.docx](AMSA_BBP_PM_04_Rev_0.docx) — Blueprint SAP PM Anglo
 - [skills/00-knowledge-base/architecture/BUEPRINT.md](../skills/00-knowledge-base/architecture/BLUEPRINT.md) — blueprint agentes
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Generado con commit: `cd83670`